

1.9: Differential Equations

1. First Order Separable Differential Equations

A differential equation is **separable** if it can be written as:

$$\frac{dy}{dx} = f(x) g(y)$$

Rearranging: $\int \frac{1}{g(y)} dy = \int f(x) dx$

Example: Solve $\frac{dy}{dx} = xy$, with $y(0) = 2$.

$$\int \frac{1}{y} dy = \int x dx \Rightarrow \ln |y| = \frac{x^2}{2} + C$$

$|y| = Ae^{x^2/2}$, so $y = Ae^{x^2/2}$ (where A can be positive or negative).

Applying $y(0) = 2$: $2 = A$. Particular solution: $y = 2e^{x^2/2}$.

Example: Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.

$$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$$

$$\arctan y = \arctan x + C$$

$$y = \tan(\arctan x + C)$$

2. First Order Linear Differential Equations

Standard form: $\frac{dy}{dx} + P(x)y = Q(x)$

Integrating factor: $\mu = e^{\int P(x) dx}$

Multiply through by μ : the left side becomes $\frac{d}{dx}(\mu y)$, giving:

$$\mu y = \int \mu Q(x) dx$$

Example: Solve $\frac{dy}{dx} + \frac{2y}{x} = x^2$.

$P(x) = 2/x$. Integrating factor: $\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$.

Multiply: $\frac{d}{dx}(x^2 y) = x^4$

Integrate: $x^2 y = \frac{x^5}{5} + C$

General solution: $y = \frac{x^3}{5} + \frac{C}{x^2}$

AQA Exam Tip: Always simplify the integrating factor $\mu = e^{\int P dx}$ before multiplying through. A common error is forgetting to multiply $Q(x)$ on the right-hand side.

3. Second Order Homogeneous Differential Equations

Form: $a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$

Auxiliary equation: $am^2 + bm + c = 0$

The general solution depends on the roots of the auxiliary equation:

Roots of auxiliary equation	General solution
Two distinct real roots m_1, m_2	$y = Ae^{m_1x} + Be^{m_2x}$
Repeated real root m	$y = (A + Bx)e^{mx}$
Complex roots $\alpha \pm \beta i$	$y = e^{\alpha x}(A \cos \beta x + B \sin \beta x)$

Example (distinct real roots): Solve $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.

Auxiliary: $m^2 - 5m + 6 = 0 \Rightarrow (m - 2)(m - 3) = 0 \Rightarrow m = 2, 3$

General solution: $y = Ae^{2x} + Be^{3x}$

Example (repeated root): Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$.

Auxiliary: $m^2 - 6m + 9 = (m - 3)^2 = 0 \Rightarrow m = 3$ (repeated)

General solution: $y = (A + Bx)e^{3x}$

Example (complex roots): Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$.

Auxiliary: $m^2 - 4m + 13 = 0 \Rightarrow m = \frac{4 \pm \sqrt{16 - 52}}{2} = 2 \pm 3i$

So $\alpha = 2, \beta = 3$.

General solution: $y = e^{2x}(A \cos 3x + B \sin 3x)$

4. Second Order Non-Homogeneous Differential Equations

Form: $ay'' + by' + cy = f(x)$

General solution: $y = y_{CF} + y_{PI}$

where y_{CF} is the complementary function (solution of the homogeneous equation) and y_{PI} is any particular integral satisfying the full equation.

Choosing a Particular Integral

$f(x)$	Trial y_{PI}
ke^{px} (where e^{px} is not in CF)	λe^{px}
ke^{px} (where e^{px} is in CF — simple root)	$\lambda x e^{px}$
ke^{px} (where e^{px} is in CF — repeated root)	$\lambda x^2 e^{px}$
$k \sin(px)$ or $k \cos(px)$	$\lambda \cos(px) + \mu \sin(px)$
Polynomial of degree n	Polynomial of degree n : $a_n x^n + \dots + a_0$
Product: polynomial $\times$ exponential	Same product form with unknown coefficients

Example: Find the general solution of $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4e^{3x}$.

CF: Auxiliary: $m^2 - 3m + 2 = (m - 1)(m - 2) = 0$, so $y_{CF} = Ae^x + Be^{2x}$.

PI: Try $y_{PI} = \lambda e^{3x}$.

$y' = 3\lambda e^{3x}, y'' = 9\lambda e^{3x}$

Substitute: $9\lambda e^{3x} - 9\lambda e^{3x} + 2\lambda e^{3x} = 4e^{3x}$

$2\lambda = 4 \Rightarrow \lambda = 2$

General solution: $y = Ae^x + Be^{2x} + 2e^{3x}$

Example: Find the general solution of $y'' - 3y' + 2y = \sin x$.

CF: $y_{CF} = Ae^x + Be^{2x}$ (as before).

PI: Try $y_{PI} = \lambda \cos x + \mu \sin x$.

$$y' = -\lambda \sin x + \mu \cos x, y'' = -\lambda \cos x - \mu \sin x$$

Substitute:

$$(-\lambda \cos x - \mu \sin x) - 3(-\lambda \sin x + \mu \cos x) + 2(\lambda \cos x + \mu \sin x) = \sin x$$

$$\text{Cosine: } -\lambda - 3\mu + 2\lambda = \lambda - 3\mu = 0$$

$$\text{Sine: } -\mu + 3\lambda + 2\mu = \mu + 3\lambda = 1$$

$$\text{From first: } \lambda = 3\mu. \text{ Substitute: } \mu + 9\mu = 1 \Rightarrow \mu = \frac{1}{10}, \lambda = \frac{3}{10}.$$

$$y = Ae^x + Be^{2x} + \frac{3}{10} \cos x + \frac{1}{10} \sin x$$

AQA Exam Tip: If your trial PI is part of the complementary function, you must multiply the trial by x (or x^2 for repeated roots). Forgetting this gives a zero coefficient and no PI.

5. Applying Initial/Boundary Conditions

Example: Solve $y'' + 4y = 8$ with $y(0) = 3, y'(0) = 0$.

CF: Auxiliary: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$. $y_{CF} = A \cos 2x + B \sin 2x$

PI: Try $y_{PI} = \lambda$ (constant). Then $y'' = 0$: $0 + 4\lambda = 8 \Rightarrow \lambda = 2$.

General solution: $y = A \cos 2x + B \sin 2x + 2$

Apply $y(0) = 3$: $A + 2 = 3 \Rightarrow A = 1$

$y' = -2A \sin 2x + 2B \cos 2x$; $y'(0) = 0$: $2B = 0 \Rightarrow B = 0$

Particular solution: $y = \cos 2x + 2$

6. Coupled Differential Equations

A system of coupled first-order equations can often be reduced to a single second-order equation.

Method for coupled ODEs: Given $\frac{dx}{dt} = ax + by$ and $\frac{dy}{dt} = cx + dy$:

1. Differentiate one equation to get $\ddot{x}$
2. Substitute the other equation to eliminate y (and $\dot{y}$)
3. Solve the resulting second-order ODE for $x(t)$
4. Back-substitute to find $y(t)$

Example: Solve $\dot{x} = 3x + 2y$, $\dot{y} = x + 2y$, with $x(0) = 1$, $y(0) = 0$.

Differentiate first equation: $\ddot{x} = 3\dot{x} + 2\dot{y} = 3\dot{x} + 2(x + 2y)$

From first equation: $y = \frac{\dot{x} - 3x}{2}$

Substitute: $\ddot{x} = 3\dot{x} + 2x + 2 \cdot 2 \cdot \frac{\dot{x} - 3x}{2} = 3\dot{x} + 2x + 2\dot{x} - 6x = 5\dot{x} - 4x$

$\ddot{x} - 5\dot{x} + 4x = 0$

Auxiliary: $m^2 - 5m + 4 = (m - 1)(m - 4) = 0$

$x = Ae^t + Be^{4t}$

$y = \frac{\dot{x} - 3x}{2} = \frac{Ae^t + 4Be^{4t} - 3Ae^t - 3Be^{4t}}{2} = -Ae^t + \frac{Be^{4t}}{2}$

Applying IC: $A + B = 1$, $-A + B/2 = 0 \Rightarrow B = 2A$, so $3A = 1 \Rightarrow A = 1/3$, $B = 2/3$.

$x = \frac{1}{3}e^t + \frac{2}{3}e^{4t}$, $y = -\frac{1}{3}e^t + \frac{1}{3}e^{4t}$

7. Equilibrium Points and Stability

For a first-order autonomous DE $\frac{dy}{dx} = f(y)$, **equilibrium points** (or fixed points) satisfy $f(y^*) = 0$.

- **Stable equilibrium:** If $f'(y^*) < 0$, small perturbations decay back to y^*
- **Unstable equilibrium:** If $f'(y^*) > 0$, perturbations grow away from y^*



8. Summary of Key Methods

Type	Method
Separable 1st order	Separate variables; integrate both sides
Linear 1st order	Integrating factor $\mu = e^{\int P dx}$
2nd order homogeneous	Auxiliary equation; CF depends on roots
2nd order non-homogeneous	General solution = CF + PI; choose PI from table
Coupled 1st order	Differentiate and eliminate to get 2nd order
Particular solution	Apply initial/boundary conditions to general solution

9. Practice Checklist

- Can you solve separable first-order ODEs?
- Can you find and apply an integrating factor?
- Can you solve second-order homogeneous ODEs for all three root types (distinct, repeated, complex)?
- Can you select the correct trial function for a particular integral and find its coefficients?
- Can you deal with the case where the trial PI is part of the CF?
- Can you find a particular solution given initial conditions?
- Can you reduce coupled first-order ODEs to a single second-order ODE?

Particular Integrals — Complete Table

For $ay'' + by' + cy = f(x)$, choose a trial particular integral based on $f(x)$:

$f(x)$	Trial PI (standard case)	Trial PI (resonance: trial is in CF)
$ke^{\alpha x}$	$\lambda e^{\alpha x}$	$\lambda x e^{\alpha x}$
$k \cos \omega x$ or $k \sin \omega x$	$\lambda \cos \omega x + \mu \sin \omega x$	$x(\lambda \cos \omega x + \mu \sin \omega x)$
kx^n (polynomial deg n)	$\lambda_n x^n + \dots + \lambda_0$ (full deg n)	$x(\lambda_n x^n + \dots + \lambda_0)$
$ke^{\alpha x} \cos \omega x$	$e^{\alpha x}(\lambda \cos \omega x + \mu \sin \omega x)$	$xe^{\alpha x}(\lambda \cos \omega x + \mu \sin \omega x)$
$ke^{\alpha x} \cdot x^n$	$e^{\alpha x}(\lambda_n x^n + \dots + \lambda_0)$	$xe^{\alpha x}(\lambda_n x^n + \dots + \lambda_0)$

Resonance rule: If the trial PI for $f(x) = ke^{\alpha x}$ (or $\sin / \cos$) is already a term in the CF, multiply the trial by x . If it is still in the CF, multiply by x^2 .

Example — resonance: Solve $y'' + 4y = 8 \sin 2x$. CF: $y_c = A \cos 2x + B \sin 2x$ (roots $\pm 2i$). Since $\sin 2x$ is in CF, resonance applies. Trial PI: $y_p = x(a \cos 2x + b \sin 2x)$.

$$y'_p = (a + 2bx) \cos 2x + (b - 2ax) \sin 2x$$

$$y''_p = (4b - 4ax) \cos 2x + (-4a - 4bx) \sin 2x$$

$$y''_p + 4y_p = 4b \cos 2x - 4a \sin 2x = 8 \sin 2x$$

So $a = -2$, $b = 0$. PI: $y_p = -2x \cos 2x$. General solution: $y = A \cos 2x + B \sin 2x - 2x \cos 2x$.

Coupled Differential Equations — Worked Example

To solve a pair of coupled first-order ODEs, use the **elimination method**:

- Express one variable in terms of the other and its derivative from the first equation.
- Differentiate and substitute into the second equation to get a single second-order ODE.

3. Solve the second-order ODE, then back-substitute for the second variable.

Example: Solve $\dot{x} = 3x - y$, $\dot{y} = x + y$. Step 1: From the first equation, $y = 3x - \dot{x}$. Step 2: Differentiate: $\dot{y} = 3\dot{x} - \ddot{x}$. Substitute into second: $3\dot{x} - \ddot{x} = x + (3x - \dot{x}) = 4x - \dot{x}$.

$$\ddot{x} - 4\dot{x} + 4x = 0$$

Step 3: Auxiliary equation: $(m - 2)^2 = 0$, so $m = 2$ (repeated root).

$$x = (At + B)e^{2t}$$

Step 4: $y = 3x - \dot{x}$. Compute $\dot{x} = (A + 2At + 2B)e^{2t}$. Then:

$$y = 3(At + B)e^{2t} - (A + 2At + 2B)e^{2t} = (At - A + B)e^{2t}$$

Check: Always verify by substituting back into *both* original equations.

Second Order ODE — Physical Interpretation

The equation $m\ddot{x} + c\dot{x} + kx = F(t)$ models a mass-spring-damper system:

- m : mass, c : damping coefficient, k : spring constant, $F(t)$: forcing function.
- Dividing by m : $\ddot{x} + 2\beta\dot{x} + \omega_0^2x = F(t)/m$ where $\omega_0 = \sqrt{k/m}$ is the natural frequency and $\beta = c/(2m)$ is the damping constant (decay rate).

Condition	Root type	Motion type
$c^2 < 4mk$ (i.e. $\beta < \omega_0$)	Complex conjugates $-\beta \pm i\omega_d$	Underdamped: damped oscillation $e^{-\beta t} \sin(\omega_d t + \phi)$
$c^2 = 4mk$	Repeated real root $-\beta$	Critically damped: fastest decay without oscillation $(A + Bt)e^{-\beta t}$
$c^2 > 4mk$	Two distinct real negative roots	Overdamped: slow exponential decay $Ae^{-\alpha_1 t} + Be^{-\alpha_2 t}$

$c = 0$	Pure imaginary $\pm i\omega_0$	Simple harmonic motion: $A \cos \omega_0 t + B \sin \omega_0 t$
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Resonance: When the forcing frequency equals the natural frequency ($F(t) = F_0 \cos \omega_0 t$, $c = 0$), the response grows without bound: $x_p = \frac{F_0 t}{2m\omega_0} \sin \omega_0 t$. This is why resonance is physically dangerous.

AQA Exam Technique — Differential Equations

Key tips:

- **Second-order CF:** Always factorise the auxiliary equation. Three cases: two distinct real roots; repeated root; complex roots $\alpha \pm \beta i$ giving $e^{\alpha x}(A \cos \beta x + B \sin \beta x)$.
- **PI selection:** Check whether the trial PI is already in the CF before choosing. If it is, multiply by x .
- **Initial conditions:** Apply to the *general solution* (CF + PI), not just the CF. Differentiate y_{GS} for the second condition.
- **Coupled equations:** After elimination, solve the resulting ODE first, then back-substitute for the other variable — do not guess its form independently.
- **Integrating factor:** For $y' + P(x)y = Q(x)$, the IF is $e^{\int P dx}$. Multiply both sides and recognise the left-hand side as $\frac{d}{dx}[y \cdot \text{IF}]$.

Type	CF	Typical PI form
$y'' + by' + cy = ke^{\alpha x}$	Depends on roots	$\lambda e^{\alpha x}$ (or $\lambda x e^{\alpha x}$ if resonant)
$y'' + by' + cy = k \sin \omega x$	Depends on roots	$\lambda \cos \omega x + \mu \sin \omega x$ (or $x \cdot$ if resonant)
$y'' + by' + cy = kx^n$	Depends on roots	Full polynomial of degree n
Euler equation $x^2 y'' + xy' + ny = 0$	Via $x = e^t$	Reduce to constant-coefficient

First-Order ODEs — Full Methods

Method 1 — Separable equations: $\frac{dy}{dx} = f(x)g(y)$.

Separate: $\frac{1}{g(y)} dy = f(x) dx$. Integrate both sides.

Example: $\frac{dy}{dx} = xy$. Separate: $\frac{1}{y} dy = x dx$.

$$\ln |y| = \frac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2} \text{ where } A = \pm e^C.$$

Method 2 — Integrating factor (linear first-order): $\frac{dy}{dx} + P(x)y = Q(x)$.

Multiply by integrating factor $\mu = e^{\int P dx}$:

$$\frac{d}{dx}(\mu y) = \mu Q \Rightarrow \mu y = \int \mu Q dx.$$

Example: $\frac{dy}{dx} + 2y = e^{-x}$.

$$\mu = e^{\int 2 dx} = e^{2x}. \text{ Multiply: } \frac{d}{dx}(e^{2x}y) = e^{2x} \cdot e^{-x} = e^x.$$

$$\text{Integrate: } e^{2x}y = e^x + C \Rightarrow y = e^{-x} + Ce^{-2x}.$$

Example 2 (non-constant P): $x\frac{dy}{dx} + y = x^2$, i.e. $\frac{dy}{dx} + \frac{1}{x}y = x$.

$$\mu = e^{\int (1/x) dx} = e^{\ln x} = x. \text{ Multiply: } \frac{d}{dx}(xy) = x^2.$$

$$\text{Integrate: } xy = \frac{x^3}{3} + C \Rightarrow y = \frac{x^2}{3} + \frac{C}{x}.$$

Second-Order ODEs — Complementary Function Cases

For $\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$, the auxiliary equation is $m^2 + bm + c = 0$.

Roots	CF	Behaviour
Real distinct: $m = \alpha, \beta$	$Ae^{\alpha x} + Be^{\beta x}$	Overdamped
Real equal: $m = \alpha$ (repeated)	$(A + Bx)e^{\alpha x}$	Critically damped
Complex: $m = p \pm qi$	$e^{px}(A \cos qx + B \sin qx)$	Underdamped / oscillatory
Pure imaginary: $m = \pm qi$	$A \cos qx + B \sin qx$	Simple harmonic / undamped

Example 1 (real distinct): $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.

AE: $m^2 - 5m + 6 = 0 \Rightarrow (m - 2)(m - 3) = 0 \Rightarrow m = 2, 3$.

CF: $y = Ae^{2x} + Be^{3x}$.

Example 2 (repeated root): $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$.

AE: $(m - 2)^2 = 0 \Rightarrow m = 2$ (repeated).

CF: $y = (A + Bx)e^{2x}$.

Example 3 (complex roots): $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$.

AE: $m^2 + 2m + 5 = 0 \Rightarrow m = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i$.

CF: $y = e^{-x}(A \cos 2x + B \sin 2x)$.

Particular Integrals — Worked Examples

Example 1 — Polynomial RHS: $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4x + 1$.

Try PI: $y_p = ax + b$. $y_p'' = 0$, $y_p' = a$.

Sub: $0 + 3a + 2(ax + b) = 4x + 1 \Rightarrow 2ax + (3a + 2b) = 4x + 1$.

$2a = 4 \Rightarrow a = 2$. $3(2) + 2b = 1 \Rightarrow b = -\frac{5}{2}$. PI: $y_p = 2x - \frac{5}{2}$.

Example 2 — Exponential RHS: $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{4x}$.

Try $y_p = \lambda e^{4x}$. Sub: $16\lambda - 12\lambda + 2\lambda = 1 \Rightarrow 6\lambda = 1 \Rightarrow \lambda = \frac{1}{6}$. Pl: $y_p = \frac{1}{6}e^{4x}$.

Example 3 — Trigonometric RHS: $\frac{d^2y}{dx^2} + 4y = \sin 3x$.

Try $y_p = A \cos 3x + B \sin 3x$. $y_p'' = -9A \cos 3x - 9B \sin 3x$.

Sub: $-9A \cos 3x - 9B \sin 3x + 4A \cos 3x + 4B \sin 3x = \sin 3x$.

$-5A = 0 \Rightarrow A = 0$. $-5B = 1 \Rightarrow B = -\frac{1}{5}$. Pl: $y_p = -\frac{1}{5} \sin 3x$.

Boundary Conditions and Initial Value Problems

Full Example: Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 6$ with $y(0) = 2$, $y'(0) = 3$.

AE: $m^2 - 4m + 3 = 0 \Rightarrow (m - 1)(m - 3) = 0$. CF: $y_c = Ae^x + Be^{3x}$.

Pl: Try $y_p = k$ (constant). $0 - 0 + 3k = 6 \Rightarrow k = 2$. Pl: $y_p = 2$.

GS: $y = Ae^x + Be^{3x} + 2$.

$y(0) = A + B + 2 = 2 \Rightarrow A + B = 0$.

$y' = Ae^x + 3Be^{3x}$. $y'(0) = A + 3B = 3$.

Solving: $B = 3/2$, $A = -3/2$.

$$y = \frac{3}{2}e^{3x} - \frac{3}{2}e^x + 2$$

Substitution Methods for ODEs

Some ODEs are not standard form; a substitution reduces them.

Bernoulli's equation: $\frac{dy}{dx} + P(x)y = Q(x)y^n$.

Substitute $v = y^{1-n}$. Then $\frac{dv}{dx} = (1 - n)y^{-n}\frac{dy}{dx}$, giving a linear equation in v .

Example: $\frac{dy}{dx} - y = y^2$. Here $n = 2$. Let $v = y^{-1}$.

$\frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$. Divide original by y^2 : $y^{-2} \frac{dy}{dx} - y^{-1} = 1$.

So $-\frac{dv}{dx} - v = 1 \Rightarrow \frac{dv}{dx} + v = -1$. Integrating factor e^x :

$\frac{d}{dx}(ve^x) = -e^x \Rightarrow ve^x = -e^x + C \Rightarrow v = -1 + Ce^{-x}$.

$y = \frac{1}{v} = \frac{1}{Ce^{-x} - 1}$.

AQA Differential Equations — Exam Technique

- **Always write GS = CF + PI** clearly. Examiners look for this structure.
- **Resonance:** If the PI trial form already appears in the CF, multiply by x . If it appears as a double root, multiply by x^2 .
- **First-order linear:** Always multiply through by IF before integrating; do not try to separate unless it genuinely is separable.
- **CF check:** Verify your CF satisfies the homogeneous equation by substituting back.
- **PI check:** Verify your PI satisfies the full equation by substituting back.
- **IVP:** Apply both initial conditions to GS to get two equations for A and B ; solve simultaneously.
- **Coupled ODEs:** Eliminate one variable by differentiating one equation and substituting the other; this gives a second-order ODE in one variable.

Modelling with Differential Equations

Common models:

- **Exponential growth/decay:** $\frac{dN}{dt} = kN$ gives $N = N_0 e^{kt}$. If $k > 0$: growth; $k < 0$: decay.
- **Newton's law of cooling:** $\frac{d\theta}{dt} = -k(\theta - \theta_{\text{amb}})$. Solution: $\theta - \theta_{\text{amb}} = Ae^{-kt}$.

- **Logistic growth:** $\frac{dN}{dt} = kN \left(1 - \frac{N}{L}\right)$ where L is the carrying capacity.
- **Harmonic oscillator:** $\frac{d^2x}{dt^2} + \omega^2x = 0$ gives $x = A \cos \omega t + B \sin \omega t$.
- **Damped oscillator:** $\frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + \omega^2x = 0$.

Example — Newton's cooling: A body at $80^\circ C$ cools in a room at $20^\circ C$. After 10 minutes it is $60^\circ C$. Find its temperature after 20 minutes.

$$\theta - 20 = Ae^{-kt}. \text{ At } t = 0: A = 60. \text{ At } t = 10: 40 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{2}{3}.$$

$$\text{At } t = 20: \theta - 20 = 60 \cdot \left(\frac{2}{3}\right)^2 = \frac{80}{3}. \theta = 20 + \frac{80}{3} \approx 46.7^\circ C.$$

Resonance — Full Worked Example

Example: Solve $\frac{d^2y}{dx^2} + 4y = \sin 2x$ (resonance case).

$$\text{CF: AE } m^2 + 4 = 0 \Rightarrow m = \pm 2i. \text{ CF: } y_c = A \cos 2x + B \sin 2x.$$

$$\text{PI: Since } \sin 2x \text{ is in CF, try } y_p = x(C \cos 2x + D \sin 2x).$$

$$y'_p = C \cos 2x + D \sin 2x + x(-2C \sin 2x + 2D \cos 2x)$$

$$y''_p = -2C \sin 2x + 2D \cos 2x + (-2C \sin 2x + 2D \cos 2x) + x(-4C \cos 2x - 4D \sin 2x) \\ = -4C \sin 2x + 4D \cos 2x - 4x(C \cos 2x + D \sin 2x)$$

$$\text{Sub into } y''_p + 4y_p: -4C \sin 2x + 4D \cos 2x = \sin 2x.$$

$$4D = 0 \Rightarrow D = 0. -4C = 1 \Rightarrow C = -\frac{1}{4}.$$

$$\text{PI: } y_p = -\frac{x}{4} \cos 2x. \text{ GS: } y = A \cos 2x + B \sin 2x - \frac{x}{4} \cos 2x.$$

Coupled ODEs — Full Worked Example

Solve: $\frac{dx}{dt} = x + 2y, \quad \frac{dy}{dt} = 3x + 2y$ with $x(0) = 1, y(0) = 0$.

$$\text{Differentiate first equation: } \ddot{x} = \dot{x} + 2\dot{y} = \dot{x} + 2(3x + 2y) = \dot{x} + 6x + 4y.$$

$$\text{From first equation: } 2y = \dot{x} - x \Rightarrow y = \frac{1}{2}(\dot{x} - x).$$

$$\text{Sub: } \ddot{x} = \dot{x} + 6x + 2(\dot{x} - x) = 3\dot{x} + 4x.$$

$\ddot{x} - 3\dot{x} - 4x = 0$. AE: $m^2 - 3m - 4 = 0 \Rightarrow (m - 4)(m + 1) = 0$. $m = 4, -1$.

$x = Ae^{4t} + Be^{-t}$. Then $y = \frac{1}{2}(\dot{x} - x) = \frac{1}{2}(4Ae^{4t} - Be^{-t} - Ae^{4t} - Be^{-t}) = \frac{3A}{2}e^{4t} - Be^{-t}$.

Conditions: $x(0) = A + B = 1$; $y(0) = \frac{3A}{2} - B = 0 \Rightarrow B = \frac{3A}{2}$.

$A + \frac{3A}{2} = 1 \Rightarrow \frac{5A}{2} = 1 \Rightarrow A = \frac{2}{5}$, $B = \frac{3}{5}$.

$x = \frac{2}{5}e^{4t} + \frac{3}{5}e^{-t}$, $y = \frac{3}{5}e^{4t} - \frac{3}{5}e^{-t}$.

Second-Order ODE Applications — SHM and Damping

A mass-spring system with damping satisfies:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

where m = mass, c = damping coefficient, k = spring constant, $F(t)$ = external force.

- **Underdamped:** $c^2 < 4mk$ — complex roots, oscillatory decay.
- **Critically damped:** $c^2 = 4mk$ — repeated real roots, fastest return to equilibrium.
- **Overdamped:** $c^2 > 4mk$ — real distinct roots, exponential decay, no oscillations.

Example (underdamped): $\ddot{x} + 2\dot{x} + 5x = 0$, $x(0) = 1$, $\dot{x}(0) = 0$.

AE: $m^2 + 2m + 5 = 0 \Rightarrow m = -1 \pm 2i$. CF: $x = e^{-t}(A \cos 2t + B \sin 2t)$.

$x(0) = A = 1$. $\dot{x} = -e^{-t}(A \cos 2t + B \sin 2t) + e^{-t}(-2A \sin 2t + 2B \cos 2t)$.

$\dot{x}(0) = -A + 2B = 0 \Rightarrow B = \frac{1}{2}$.

$x = e^{-t}(\cos 2t + \frac{1}{2} \sin 2t)$.

Key Formulae Summary — Differential Equations

Equation type	Method	Key step
Separable: $\frac{dy}{dx} = f(x)g(y)$	Separate variables	$\int \frac{1}{g} dy = \int f dx$

Linear 1st order: $y' + Py = Q$	Integrating factor	$\mu = e^{\int P dx}$
2nd order homogeneous	Auxiliary equation	Solve $m^2 + bm + c = 0$
2nd order inhomogeneous	CF + PI	Match PI form to RHS
Coupled 1st order	Elimination	Differentiate one, sub other

Summary of PI forms:

RHS	PI trial form	If resonance
$ke^{\alpha x}$	$\lambda e^{\alpha x}$	$\lambda x e^{\alpha x}$
$k \cos \omega x$ or $k \sin \omega x$	$\lambda \cos \omega x + \mu \sin \omega x$	Multiply by x
kx^n	$\lambda_n x^n + \dots + \lambda_1 x + \lambda_0$	N/A (unless $c = 0$)
$ke^{\alpha x} \cos \omega x$	$e^{\alpha x} (\lambda \cos \omega x + \mu \sin \omega x)$	Multiply by x

Euler–Cauchy Equations

The Euler–Cauchy equation $x^2 \frac{d^2 y}{dx^2} + px \frac{dy}{dx} + qy = 0$ is solved by the substitution $x = e^t$ (i.e. $t = \ln x$), which transforms it into a constant-coefficient ODE in t .

Example: Solve $x^2 y'' - 2y = 0$.

With $x = e^t$: $x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} - \frac{dy}{dt}$.

Equation becomes: $\frac{d^2 y}{dt^2} - \frac{dy}{dt} - 2y = 0$.

AE: $m^2 - m - 2 = 0 \Rightarrow (m - 2)(m + 1) = 0 \Rightarrow m = 2, -1$.

GS in t : $y = Ae^{2t} + Be^{-t}$. Back-substitute $t = \ln x$: $y = Ax^2 + Bx^{-1}$.

Note for AQA: Euler–Cauchy equations may appear in AQA Further Pure papers. The substitution $x = e^t$ reduces them to constant-coefficient form. Try a power-law solution $y = x^m$ directly: substituting gives the indicial equation $m(m - 1) + pm + q = 0$.

Superposition Principle

For a linear ODE $Ly = f(x)$, if y_1 is a solution when $f = f_1$ and y_2 when $f = f_2$, then $y_1 + y_2$ is a solution when $f = f_1 + f_2$.

This allows complex RHS functions to be split into simpler parts, each solved separately.

Example: To solve $y'' + y = \sin x + e^x$, find PI for $\sin x$ and PI for e^x separately, then add.

PI₁ for $\sin x$: since $m = \pm i$ are roots of AE, there is resonance. Try $y_{p1} = x(A \cos x + B \sin x)$.

PI₂ for e^x : try $y_{p2} = \lambda e^x$. $\lambda e^x + \lambda e^x = e^x \Rightarrow \lambda = \frac{1}{2}$.

GS: $y = A \cos x + B \sin x + y_{p1} + \frac{1}{2}e^x$.